VLM benchmark

4 vision models, each asked the question three different ways, on the same 654 image crops. The setup currently in use comes out on top and stays unchanged.

Findings

1. **No configuration beats the one in use.** GLM-4.6V-Flash-9B with the cautious wording keeps 55 of 73 objects, raises no false alarm, and 90 % of what it reports is real. None of the other 11 runs improves on that.
2. **The higher scores in the table are not real.** 6 runs report more objects, but they answer yes to between 29 and 89 % of every crop shown to them. They would raise an alarm on most clean trams. The benchmark implies a correct rate of about 11 %.
3. **Rewording the question changes nothing here.** On the model in use the three wordings give 55, 55 and 56 objects. It does help a model that is too cautious to begin with: on InternVL the same rewording goes from 39 to 48 objects with no false alarm.
4. **Choosing a different off-the-shelf model has run its course.** The 18 objects still missed are not recovered by any of these models at an acceptable false-alarm rate. Training a model on our own images is the next step.

Method

The image crops were produced once, by the best configuration from the models note, and reused unchanged in all 12 runs. Every model therefore saw exactly the same 654 crops of the same 68 cases, so any difference in the results comes from the model or the wording and nothing else.

Three wordings of the same question were tried, differing only in how cautious the model is told to be:

Conservative The wording in use today. It spells out when to answer no: a change of light, a person, a reflection on metal. It also tells the model to answer no whenever it is unsure.

Lenient The original wording. It lists what counts as an anomaly and the three cases that do not count, and stops there.

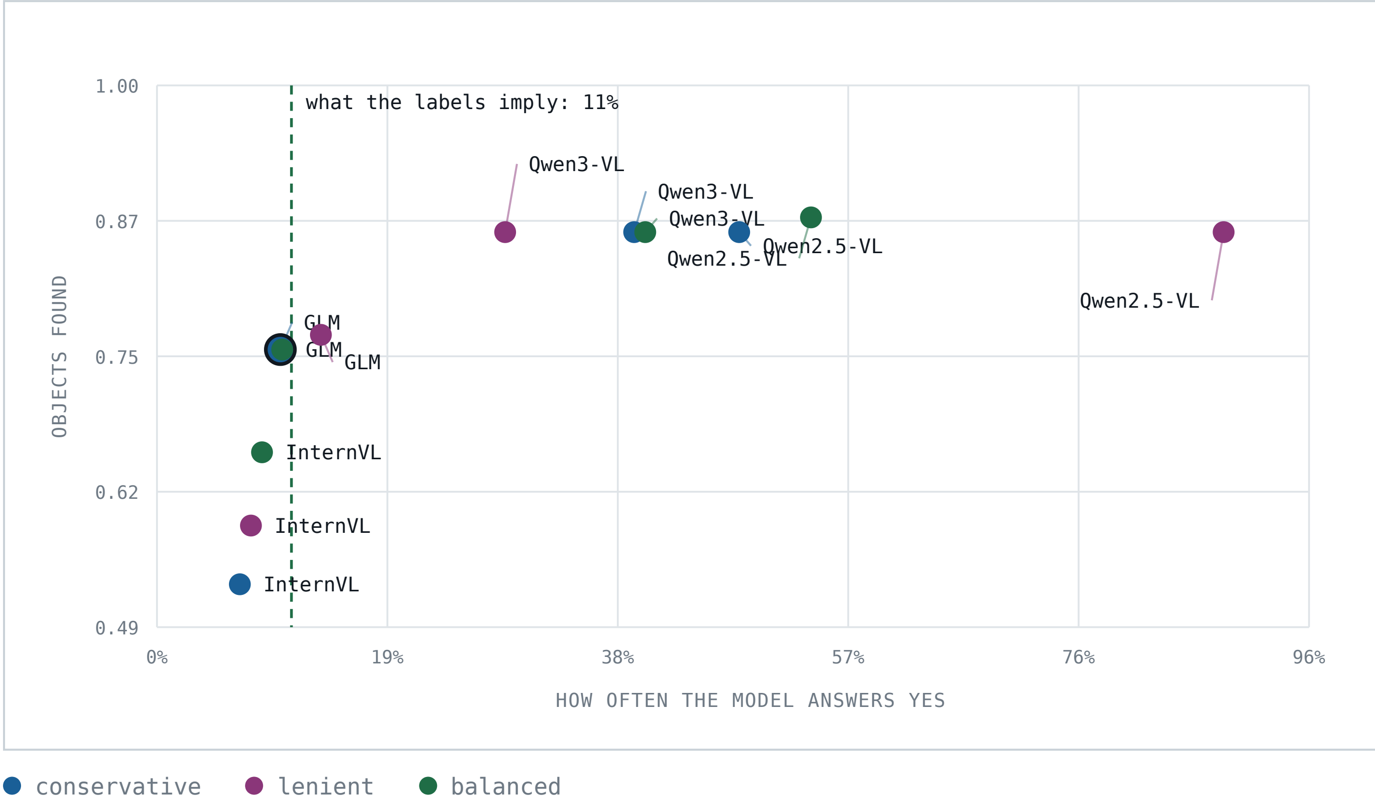
Balanced Written for this comparison. It keeps every reason to answer no from the cautious version, but drops the instruction to answer no when unsure, and says that an object can be small or half hidden and still count.

Two figures are reported for every run. How many of the 73 labelled objects survive the judge, and how often the model answers yes at all. The second one matters: the benchmark holds 73 real anomalies among 654 crops, so a correctly calibrated model should answer yes to about 11 % of them. A model well above that is not finding more, it is refusing less.

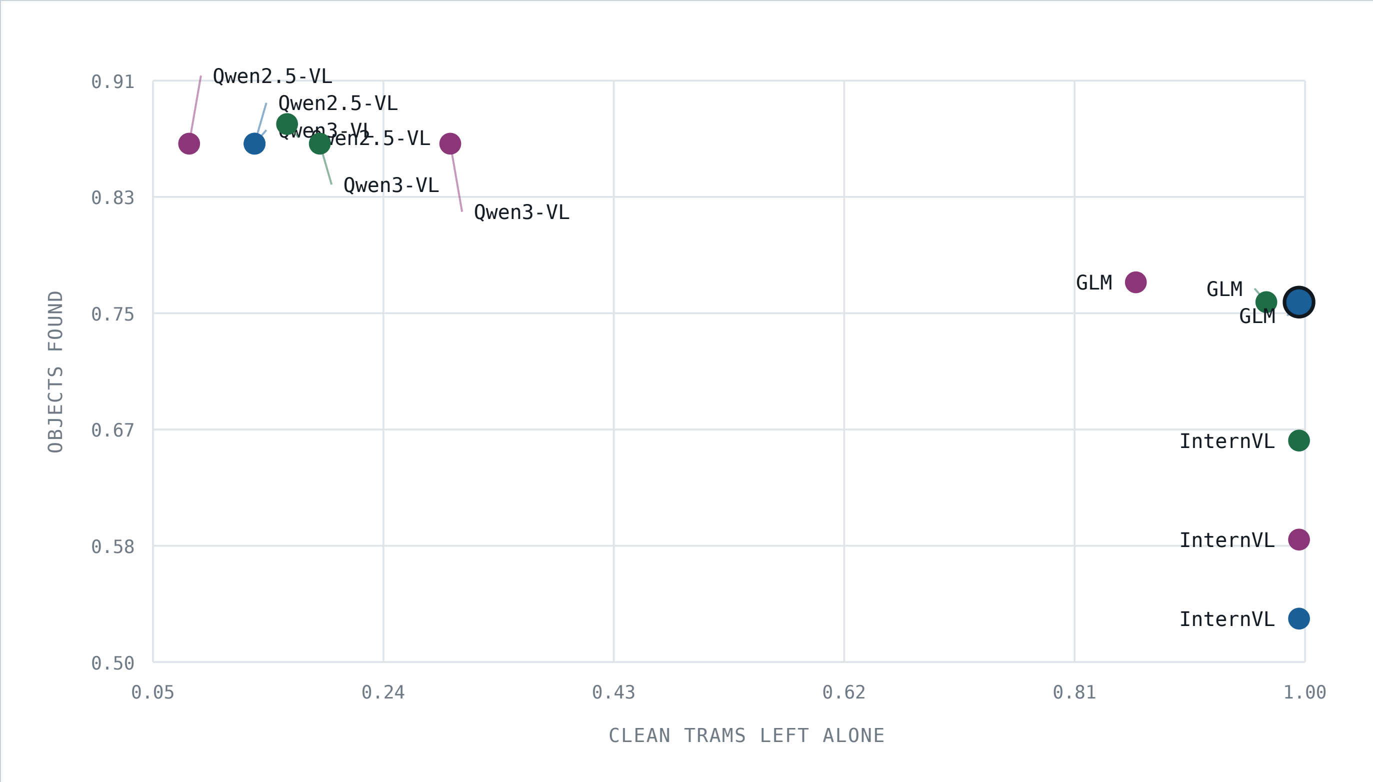
Results

JUDGE	CONSERVATIVE	LENIENT	BALANCED
Qwen2.5-VL-7B	63/73 yes 48% · right 22% · **alarms 86%**	63/73 yes 89% · right 12% · **alarms 92%**	**64/73** yes 54% · right 20% · **alarms 84%**
Qwen3-VL-8B	63/73 yes 40% · right 27% · **alarms 86%**	63/73 yes 29% · right 38% · **alarms 70%**	63/73 yes 41% · right 27% · **alarms 81%**
GLM-4.6V-Flash-9B	55/73 yes 10% · right 90%	56/73 yes 14% · right 72% · **alarms 14%**	55/73 yes 10% · right 88% · **alarms 3%**
InternVL3.5-8B	39/73 yes 7% · right 93%	43/73 yes 8% · right 92%	48/73 yes 9% · right 91%

Objects kept out of 73, then how often the model said yes, how many of those answers were right, and the share of clean trams it would have raised an alarm on. That last figure appears only where a run raised any.



The dashed line is the rate a correctly calibrated model should answer yes at. The score rises to the right because a model that rarely says no keeps everything it is given. The circled point is the setup in use.



The same runs against false alarms. Higher is better, further right is safer. Four runs never raise a false alarm, and the one in use is the best of those.

Limits

- **One set of crops.** A wording suited to these tight crops may not suit the much larger regions the pixel-difference detector produces. The result is which model for these crops, not which model in general.
- **The false-alarm figure rests on few clean images**, and the 3333* clean frames come from the same recordings the filtering model was trained on. A run that raises no alarm here is not proven safe.
- **Three wordings is not a search of the prompt space.** They differ on one axis only, which is what makes them comparable.
- **Nothing was fine-tuned.** Prompts cost minutes to try, so they went first. The groundwork for training on our own images already exists in the project and has never been run.
- **Side note.** `qwen3-vl:8b-instruct` is still the default model name written in `vlm_05_reference_diff.py`, although the benchmark has always been run with GLM. On these crops that default is unusable.

* 3333 is a placeholder, not the tram's real fleet number - the vehicle number of the 2026-08-11 capture is unknown. It was called 39T before, but 39T is the Škoda type, which tram 1760 shares.